latents verification matrix

Package 1.0.0. Built from saved evidence on 2026-10-29; nothing was refitted. 35,806 current checks (3,446,037 values) against references other than latents itself.

A **check** is one harness quantity (a whole output column counts once; *values* counts every number inside it) or one testthat expectation. **Current** means produced on 2026-10-28 or 2026-10-29 by the R code shipped in 0.98.0 and 0.98.1 (unchanged in 1.0.0). Results from 0.96.1 (2026-10-20) are shown in grey and do not count toward strength.

Legend

Strength	One **+** per *source*: an independent program or package (another R package, Mplus, Latent GOLD, Stata, Python) whose results agree on the current code for a model with two or more classes. One **P** per published paper whose printed numbers are reproduced. **++++ PP** is four sources and two papers. Colour: **0** no source, **+** one, **++** two, **+++** three or more.
Not counted	Agreement on a one-class reduction only (one class equals lm(), glm.nb(), polr(), lme4, a linear mixed model), shown as "(1 class)"; our own independent code (numDeriv, dense likelihoods, the kits' reference implementations); internal checks (invariance, seed stability, refusal parity); results last run under 0.96.1.
Sources, papers	The number of + and of P.
Datasets	Distinct datasets on which the function was compared with a source or a paper on current code, for models with two or more classes: the harness's datasets, the kits' datasets (kit-datasets.csv) and the simulated datasets of the testthat kits that the other program ran (test-datasets.csv). A dataset used by several kits counts once. Tests that build their own small data inline are not counted, so this is a lower bound.
Checks passed	Current checks against every reference other than latents itself: one harness quantity (a whole output column), one kit quantity or one testthat expectation. *Values* counts the numbers inside the harness checks.
Failed	Current checks that disagree without a registered, explained difference (each kit's known-differences.csv).
Grid cells	**bold**: current checks on mixtures; (1 class): one-class reductions only; *grey italic*: last run under 0.96.1 or no saved result; candidate: found, not yet wired in; *: a test that checks several functions or references at once (counted once in each function's total).

Strength by function

function	strength	sources	papers	datasets	checks passed	failed	values (harness)	independent programs and packages	published papers	to strengthen
lpa()	**++++++++ PPP**	8	3	163	8,049	0	124,207	Mplus, Latent GOLD, Stata 19 [SEM] Example 52g (gsem), mclust, Rmixmod, mixtools, EMCluster, tidyLPA	Masyn (2013), Oxford Handbook of Quantitative Methods, Table 25.12, Scrucca et al. (2023) mclust book, Houle et al. (2026)	
multilpa()	**+++++++**	7	0	91	7,367	0	195,348	Mplus, Latent GOLD, depmixS4, MASS, VGAM, mclust, stats lm/glm		one-class checks only, not counted: sandwich; no published paper found
lca()	**+++++++++ PPPPPPP**	9	7	161	13,072	0	256,296	Mplus, Latent GOLD, Stata 19 [SEM] Example 50g (gsem), poLCA, depmixS4, glca, BayesLCA, randomLCA, multilevLCA	Goodman (2002), Kleinman and Horton (2011), SAS and R blog, Example 8.21, Linzer & Lewis (2011), Dayton (1998), Zhou & Lange (2010), McCutcheon (1987), Agresti & Lang (1993)	
multilca()	**++++ P**	4	1	25	5,376	0	2,531,038	Mplus, Latent GOLD, glca, multilevLCA	Lyrvall et al. (2025)	
lta()	**+++++ PPP**	5	3	31	2,456	0	255,596	Latent GOLD, Mplus, depmixS4, LMest, seqHMM	Pennoni, Pandolfi and Bartolucci (2024), R Journal RJ-2024-036, section 3.3, Muthén & Asparouhov (2022), Bartolucci, Pandolfi and Pennoni (2017), JSS 81(4), section 3.1	
mixture_regression()	**+++++ PPP**	5	3	21	719	0	38,512	Latent GOLD, Mplus, flexmix, mixtools, MoEClust	SAS/STAT 14.2 PROC FMM Example 40.3 (Wang et al. 1996 model), Benaglia et al.	one-class checks only, not counted: MASS,

function	strength	sources	papers	datasets	checks passed	failed	values (harness)	independent programs and packages	published papers	to strengthen
									(2009) JSS 32(6), CO2data, Grun and Leisch (2008) JSS 28(4), betablocker	stats lm/glm, sandwich
mixture_regression(random=)	+++ P	3	1	14	827	0	40,748	Mplus 9 Demo (JStats fixtures), Latent GOLD, lcmm	Proust-Lima, Philipps and Liquet (2017), JSS 78(2), section 6	one-class checks only, not counted: nlme, lme4
three_step() / r3step()	+++++	5	0	77	2,350	0	927	Latent GOLD, Mplus, depmixS4, sandwich, tidySEM		1 test file(s) never saved a result; first paper from: Bakk & Kuha (2021) BJMSP
bootstrap_lrt() / VLMR	+++	3	0	63	5,485	0	8	Mplus, Latent GOLD, mclust		1 test file(s) never saved a result; first paper from: Masyn (2013) Table 25.12 and Stata lcstats
enumerate_*() / compare_models()	++++	4	0	10	245	0	196	Mplus, Latent GOLD, mclust, poLCA		no published paper found
pool_imputations()	+++	3	0	2	136	0	3,161	mice, mitml, mitools		no published paper found

Reference by function

Each cell: the current checks of a function against a reference (styles in the legend). Internal checks (invariance, seed stability, refusal parity, identities) are in the evidence pages, not in this grid.

reference	lpa()	multilpa()	lca()	multilca()	lta()	mixture_ regression()	growth (random =)	three_step() r3step()	bootstrap_ lrt() VLMR	enumerate_* () compare_ models()	pool_ imputations()
strength (+ source, P paper)	++++++++ PPP	+++++++	+++++++++ PPPPPPP	++++ P	+++++ PPP	+++++ PPP	+++ P	+++++	+++	++++	+++
datasets tested	**163**	**91**	**161**	**25**	**31**	**21**	**14**	**77**	**63**	**10**	**2**
COMMERCIAL PROGRAM											
Mplus	**4,130***	**3,183***	**9,903***	**3,605***	**40***	**6**		**46**	**5,477***	**49**	
Latent GOLD	**2,038***	**2,464***	**7,593***	**650***	**149***	**120**	**91**	**231***	**5,462***	**24**	
Mplus 9 Demo (JStats fixtures)							**134**				
Stata 19 [SEM] Example 52g (gsem)	**74**										
Stata 19 [SEM] Example 50g (gsem)			**22**								
IMMERSE lca-bookdown (Mplus 8.11)					candidate						
IMMERSE Nylund-Gibson et al. (2019)								candidate			
Mplus 9 Demo LCGA/GMM (JStats)							candidate				
Mplus Demo TECH11/TECH14 (JStats)									candidate		
Mplus UG ex7.6/7.7/7.8 (JStats)			candidate								
Mplus UG ex8.12/8.15/10.13					candidate						
Mplus UG ex8.9							candidate				
Mplus Web Talk 8 BCH (JStats)								candidate			
Stata 19 [FMM] Examples 1a/1b/2						candidate					
Stata 19 [SEM] Example 52g	candidate										
Stata 19 [SEM] Examples 50g/51g			candidate								
R PACKAGE											
mclust	**3,465***	**8**							**8**	**120**	
depmixS4		**290**	**52**		**1,822***			**157**			
poLCA			**1,925***							**36**	
glca			**52**	**552**							
multilevLCA			**25**	**549**							
lcmm							**535**				
flexmix						**482**					
LMest					**395***						
MASS		**160**				9 (1 class)					
sandwich		95 (1 class)				1 (1 class)		**56**			
VGAM		**83**									
Rmixmod	**75**										
mixtools	**50**					**20**					
mice											**62**
tidySEM								**43**			
BayesLCA			**33**								
EMCluster	**30**										
randomLCA			**30**								

reference	lpa()	multilpa()	lca()	multilca()	lta()	mixture_ regression()	growth (random =)	three_step() r3step()	bootstrap_ lrt() VLMR	enumerate_* () compare_ models()	pool_ imputations()
mitml											**20**
MoEClust						**20**					
mitools											**16**
stats lm/glm		**3**				8 (1 class)					
tidyLPA	**11**										
seqHMM					**8**						
nlme							5 (1 class)				
lme4							4 (1 class)				
mclust bootstrap LRT (2016 paper; vignette)									candidate		
mclust vignette (Scrucca et al.), diabetes									*no result*		
nlme corAR1/corCAR1 identity							candidate				
randomLCA (Beath 2017)									candidate		
Rmixmod (Lebret et al. 2015)	candidate										
PYTHON PACKAGE											
StepMix (Morin et al. 2025)	candidate		candidate								
StepMix (Morin et al. 2025) Table 9								candidate			
PUBLISHED											
Masyn (2013), Oxford Handbook of Quantitative Methods, Table 25.12	**49**										
Proust-Lima, Philipps and Liquet (2017), JSS 78(2), section 6							**19**				
SAS/STAT 14.2 PROC FMM Example 40.3 (Wang et al. 1996 model)						**14**					
Goodman (2002)			**12**								
Pennoni, Pandolfi and Bartolucci (2024), R Journal RJ-2024-036, section 3.3					**12**						
Kleinman and Horton (2011), SAS and R blog, Example 8.21			**11**								
Linzer & Lewis (2011)			**9**								
Muthén & Asparouhov (2022)					**8**						
Dayton (1998)			**6**								
Zhou & Lange (2010)			**6**								
McCutcheon (1987)			**5**								
Bartolucci, Pandolfi and Pennoni (2017), JSS 81(4), section 3.1					**4**						
Benaglia et al. (2009) JSS 32(6), CO2data						**4**					
Agresti & Lang (1993)			**3**								
Lyrvall et al. (2025)				**3**							
Scrucca et al. (2023) mclust book	**3**										
Grun and Leisch (2008) JSS 28(4), betablocker						**2**					
Houle et al. (2026)	**2**										
Bakk & Kuha (2021) BJMSP								candidate			
Deb & Trivedi (1997)						candidate					
Masyn (2013)	candidate										

reference	lpa()	multilpa()	lca()	multilca()	lta()	mixture_ regression()	growth (random =)	three_step() r3step()	bootstrap_ lrt() VLMR	enumerate_* () compare_ models()	pool_ imputations()
Masyn (2013) Table 25.12 and Stata lcstats									candidate		
Scrucca et al. (2016)	candidate										
INDEPENDENT CODE											
ML three-step reference (reference.R)								**1,793**			
base-R likelihood + numDeriv	**125**	**321**		**17**				**18**			**16**
numDeriv / dense likelihood	**103***	**267***			**29**	**9**	**19**				
mvtnorm dense density		**178**									
reference optimizer			**27**								
Rmpfr exact arithmetic						**3**					
sandwich (glm composition)						**3**					
sandwich (lm composition)						**1**					
current checks, all references	**8,049**	**7,367**	**13,072**	**5,376**	**2,456**	**719**	**827**	**2,350**	**5,485**	**245**	**136**

Evidence by function

lpa()

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Mplus	commercial program	current	4,130	141				*
Latent GOLD	commercial program	current	2,038	145	4,938			*
Stata 19 [SEM] Example 52g (gsem)	commercial program	current	74	1	74			
mclust	R package	current	3,465	75	84,144			*
Rmixmod	R package	current	75	6	7,898			
mixtools	R package	current	50	6	5,022			
EMCluster	R package	current	30	6	3,128			
tidyLPA	R package	current	11					
Masyn (2013), Oxford Handbook of Quantitative Methods, Table 25.12	published	current	49	1	49			
Scrucca et al. (2023) mclust book	published	current	3	1	3			
Houle et al. (2026)	published	current	2					
base-R likelihood + numDeriv	independent code	current	125		3,421			
numDeriv / dense likelihood	independent code	current	103					*
invariance	internal	current	352		15,520			
vignette claims	internal	current	18					
refusal parity	internal	current	7		7			
internal identity	internal	current	3					
seed stability	internal	current	3		3			

Candidates not yet wired in

reference	kind	data	published numbers	checked
Masyn (2013)	published	Reaven-Miller diabetes /10 (Stata gsem_lca2.dta, n = 145)	Table 25.12: LL, BIC, adj. LMR for K = 1-5 x EEI/VVI/EEE/VVV (VVV-3 LL -1536.64); Table 25.14 entropy .88, AvePP, OCC	verified
Stata 19 [SEM] Example 52g	commercial program	same as Masyn (2013)	EEI-2 LL -1702.5542, VVV-3 LL -1536.6409, every parameter with SE	verified
StepMix (Morin et al. 2025)	Python package	iris; mclust::banknote	iris VVV-3 LL -180.185; banknote VVI-2 with covariate LL -771.669	verified
Scrucca et al. (2016)	published	old mclust::diabetes; gclus::wine	14 structures x G = 1-9 BIC grid; wine EVE-3 BIC -6873.257; BLRT statistics	verified
Rmixmod (Lebret et al. 2015)	R package	geyser	K = 3 LL -1131.0738, BIC 2312.5998	verified

multilpa()

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Mplus	commercial program	current	3,183	56				*
Latent GOLD	commercial program	current	2,464	85	135,223			*
depmixS4	R package	current	290	3	13,772			
MASS	R package	current	160	2	9,032			
sandwich	R package	one class only	95		3,474			
VGAM	R package	current	83	1	5,693			
mclust	R package	current	8					
stats lm/glm	R package	current	3					
base-R likelihood + numDeriv	independent code	current	321		17,495			
numDeriv / dense likelihood	independent code	current	267					*
mvtnorm dense density	independent code	current	178					
invariance	internal	current	279		10,649			
simulation recovery	internal	current	21					
internal identity	internal	current	10					
refusal parity	internal	current	10		10			

Candidates not yet wired in

reference	kind	data	published numbers	checked
singleton-group identity	internal	any lpa() reference	groups of size one: multilpa() equals lpa()	identity

lca()

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Mplus	commercial program	current	9,903	136				*
Latent GOLD	commercial program	current	7,593	139	5,104			*
Stata 19 [SEM] Example 50g (gsem)	commercial program	current	22	1	22			
poLCA	R package	current	1,925	70	173,916			*
depmixS4	R package	current	52	9	30,914			
glca	R package	current	52	9	30,914			
BayesLCA	R package	current	33	8	305			
randomLCA	R package	current	30	7	277			
multilevLCA	R package	current	25	8	260			
Goodman (2002)	published	current	12	1	12			
Kleinman and Horton (2011), SAS and R blog, Example 8.21	published	current	11	1	11			
Linzer & Lewis (2011)	published	current	9	1	9			
Dayton (1998)	published	current	6	1	6			
Zhou & Lange (2010)	published	current	6	1	6			
McCutcheon (1987)	published	current	5	1	5			
Agresti & Lang (1993)	published	current	3	1	3			
reference optimizer	independent code	current	27					
invariance	internal	current	235		14,524			
search protocol	internal	current	29					
verification protocol	internal	current	21					
vignette claims	internal	current	16					
refusal parity	internal	current	5		5			
seed stability	internal	current	3		3			
internal identity	internal	current	2					

Candidates not yet wired in

reference	kind	data	published numbers	checked
Stata 19 [SEM] Examples 50g/51g	commercial program	Stouffer-Toby (gsem_lca1.dta, n = 216)	2 classes LL -504.46767 with SEs; entropy and LMR for K = 1-3	verified
Mplus UG ex7.6/7.7/7.8 (JStats)	commercial program	UG data in ../JStats/validation/r-reference	ex7.6 LL -19214.480; ex7.7/7.8 LL -21408.482, estimates with SEs	verified
StepMix (Morin et al. 2025)	Python package	data_bakk_covariate(2000) exported from Python	LL -5192.049 (StepMix = poLCA = Mplus)	verified

multilca()

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Mplus	commercial program	current	3,605	16	395,792			*
Latent GOLD	commercial program	current	650	16	15			*
glca	R package	current	552	4	72,424			
multilevLCA	R package	current	549	4	2,056,646			
Lyrvall et al. (2025)	published	current	3	1	3			
base-R likelihood + numDeriv	independent code	current	17		6,153			
seed stability	internal	current	3		3			
refusal parity	internal	current	2		2			

Candidates not yet wired in

reference	kind	data	published numbers	checked
singleton-group identity	internal	any lca() reference	groups of size one: multilca() equals lca()	identity

lta()

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Latent GOLD	commercial program	current	149	9	9,731			*
Mplus	commercial program	current	40					*
depmixS4	R package	current	1,822	17	194,830			*
LMest	R package	current	395	5	50,963			*
seqHMM	R package	current	8	1	45			
Pennoni, Pandolfi and Bartolucci (2024), R Journal RJ-2024-036, section 3.3	published	current	12	1	12			
Muthén & Asparouhov (2022)	published	current	8					
Bartolucci, Pandolfi and Pennoni (2017), JSS 81(4), section 3.1	published	current	4	1	4			
numDeriv / dense likelihood	independent code	current	29					
refusal parity	internal	current	11		11			

Candidates not yet wired in

reference	kind	data	published numbers	checked
IMMERSE lca-bookdown (Mplus 8.11)	commercial program	LSAY lsay_lta.csv	invariant LL -14542.768; occasion-specific LL -14531.997	verified
Mplus UG ex8.12/8.15/10.13	commercial program	posted with the examples	HMM LL -13427.811; mover-stayer LL -19504.367; two-level LTA LL -20716.373	online

mixture_regression()

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Latent GOLD	commercial program	current	120	7	4,918			
Mplus	commercial program	current	6					
flexmix	R package	current	482	10	31,118			
mixtools	R package	current	20	4	970			
MoEClust	R package	current	20	4	1,470			
MASS	R package	one class only	9					
stats lm/glm	R package	one class only	8					
sandwich	R package	one class only	1					
SAS/STAT 14.2 PROC FMM Example 40.3 (Wang et al. 1996 model)	published	current	14	1	14			
Benaglia et al. (2009) JSS 32(6), CO2data	published	current	4	1	4			
Grun and Leisch (2008) JSS 28(4), betablocker	published	current	2	1	2			
numDeriv / dense likelihood	independent code	current	9					
Rmpfr exact arithmetic	independent code	current	3					
sandwich (glm composition)	independent code	current	3					
sandwich (lm composition)	independent code	current	1					
invariance	internal	current	9		9			
refusal parity	internal	current	7		7			
internal identity	internal	current	1					

Candidates not yet wired in

reference	kind	data	published numbers	checked
Stata 19 [FMM] Examples 1a/1b/2	commercial program	mus03sub.dta, gsem_mixture.dta	3-class Gaussian LL -4727.6738; lcprob LL -4712.3871; Poisson LL -11502.686	verified
Deb & Trivedi (1997)	published	AER::NMES1988	2-class negative binomial (tables not opened)	online

mixture_regression(random =)

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Mplus 9 Demo (JStats fixtures)	commercial program	current	134	4	134			
Latent GOLD	commercial program	current	91	3	5,688			
lcmm	R package	current	535	7	34,083			
nlme	R package	one class only	5					
lme4	R package	one class only	4					
Proust-Lima, Philipps and Liquet (2017), JSS 78(2), section 6	published	current	19	1	19			
numDeriv / dense likelihood	independent code	current	19					
unit-scale refit	internal	current	10		824			
internal identity	internal	current	9					
unit invariance	internal	current	1					

Candidates not yet wired in

reference	kind	data	published numbers	checked
Mplus 9 Demo LCGA/GMM (JStats)	commercial program	../JStats/tests/fixtures/mplus (lcga-ml; gmm-ml; runner-*)	LCGA K = 2 LL -2091.416; GMM K = 2 LL -2727.520	verified
Mplus UG ex8.9	commercial program	posted with the example	binary LCGA LL -1272.307	online
nlme corAR1/corCAR1 identity	R package	any growth data	one class with AR/Brownian residuals equals nlme::lme	identity

three_step() / r3step()

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Latent GOLD	commercial program	current	231	74	30			*
Mplus	commercial program	current	46	70				
depmixS4	R package	current	157	3	638			
sandwich	R package	current	56	2	106			
tidySEM	R package	current	43	2	93			
ML three-step reference (reference.R)	independent code	current	1,793					
base-R likelihood + numDeriv	independent code	current	18		60			
internal identity	internal	current	6					
runtime guard	internal	no saved result	0				1	

Candidates not yet wired in

reference	kind	data	published numbers	checked
StepMix (Morin et al. 2025) Table 9	Python package	GSS 1976-77 via gssr	BCH 26.71/36.81/44.68; ML LL -18754.72	verified
Bakk & Kuha (2021) BJMSP	published	GSS 1976-77 (ICPSR)	BCH 26.65/37.06/44.70; ML 20.63/44.94/65.67	online
Mplus Web Talk 8 BCH (JStats)	commercial program	mplus-webtalk8-distal.dat	LL -3634.865; BCH means with SEs	verified
IMMERSE Nylund-Gibson et al. (2019)	commercial program	LSAY lsay_df.csv	BCH step 3 LL -12744.043; ML step 3 LL -12742.252	verified

bootstrap_lrt() / VLMR

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Mplus	commercial program	current	5,477	62				*
Latent GOLD	commercial program	current	5,462	62				*
mclust	R package	current	8	1	8			
mclust vignette (Scrucca et al.), diabetes	R package	no saved result	0					
runtime guard	internal	no saved result	0				1	

Candidates not yet wired in

reference	kind	data	published numbers	checked
mclust bootstrap LRT (2016 paper; vignette)	R package	mclust::diabetes (two versions)	statistics 361.186445 / 114.703559 / 7.437806	verified
randomLCA (Beath 2017)	R package	randomLCA::myocardial	bootstrap LRT p = .001	verified
Masyn (2013) Table 25.12 and Stata lcstats	published	Reaven-Miller /10	adjusted LMR p-values	verified
Mplus Demo TECH11/TECH14 (JStats)	commercial program	mplus-lca-tech11/tech14-s1/s2; lca-boundary-tech11	TECH11/TECH14 for K = 2, 3	verified

enumerate_*() / compare_models()

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
Mplus	commercial program	current	49	2				
Latent GOLD	commercial program	current	24	2	24			
mclust	R package	current	120	3	120			
poLCA	R package	current	36	3	36			
refusal parity	internal	current	16		16			

pool_imputations()

reference	kind	status	current checks	datasets	values	0.96.1 checks	files without result	shared
mice	R package	current	62	1	1,498			
mitml	R package	current	20	2	425			
mitools	R package	current	16	2	340			
base-R likelihood + numDeriv	independent code	current	16		454			
direct multilpa() call	internal	current	14		444			
internal identity	internal	current	8					

Inputs

input	run	package version	used
validation/release-fixes-2026-10-28/equivalence/report.csv	2026-10-28	0.98.0	current results
validation/release-fixes-2026-10-28/release/external-tests-consolidated.csv	2026-10-28	0.98.0	current results
validation/release-fixes-2026-10-28/release/retained-references-results.csv	2026-10-28	0.98.0	current results
validation/matrix-refresh-2026-10-28/equivalence-results.csv	2026-10-29	0.98.1	current results
validation/release-0.96.1-2026-10-20/outer-final/cases.csv	2026-10-20	0.96.1	only for files without a current result

Attribution rules: harness-functions.csv, harness-references.csv and test-attribution.csv beside build.R. lca(): every indicator categorical, one level; multilca(): the same with groups; lpa(): every indicator Gaussian, one level; multilpa(): everything else, including count and ordinal indicators and tests that call the engine directly.